

ANDRIJANA MIKOVIĆ

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Work experience

GDI SOLUTIONS

SOFTWARE ENGINEER I

April 2026 - Present
Vue.js, Node.js, PostgreSQL, Python/ArcPy, legacy Java/Vuetify
– Contributed to migrating GDi Ensemble Smart Portal's backend from Java to Node.js and consolidating its database layer on PostgreSQL, moving off the legacy MSSQL/Oracle/PostgreSQL multi-database setup — while continuing to maintain and build custom components for customers still on the legacy Java/Vuetify version.
– Implemented JWT authentication and role/privilege-based authorization, and built REST endpoints and business logic used across the platform.
– Leading development of the team's in-house UI component library — a from-scratch replacement for Vuetify — having started key parts of it myself and taken on a growing role in architecture and design decisions.
– Built a GP (geoprocessing) tool with Python and ArcPy for automated spatial data export and analysis.
– Earlier, built an AI-assisted configuration generator using a self-hosted Llama LLM with custom prompt formatting tailored to the platform's infrastructure.

JUNIOR SOFTWARE ENGINEER

April 2024 – March 2026 [Java, Vue.js, ArcGIS API, Databases]
– Developed new platform features with Vue 2.7 and Vuetify, and built reusable map widgets (file upload, import/export, forms, tables, filters).
– Implemented Java backend functionality — REST endpoints, database calls, and proxy-related logic — and added compatibility for Oracle and PostgreSQL alongside the primary MSSQL database.
– Worked with Vuex and Pinia for state management and helped define the architecture for new modules together with the team.
– Participated in deployments, production debugging, and support across multiple enterprise customer projects.

MICROSOFT DEVELOPMENT CENTER SERBIA - INTERNSHIP

March 2022 – May 2022 [React.js, D3.js]
I was a part of the Maps team, and worked with them on the existing web application for modifying maps data.

Education

SCHOOL OF ELECTRICAL ENGINEERING UNIVERSITY OF BELGRADE

Software Engineering

2023 – 2025
Master Academic Studies, took courses like Social Network Analysis, big data, cyber security etc. Thesis: Analysis of trends in world cinematography in the early 21st century.

Computer Engineering and Information Technology

2019 – 2023
Bachelor studies. A strong foundation in computer engineering, programming, and information technology.

FIRST GRAMMAR SCHOOL KRAGUJEVAC

2013 - 2019
I went to class for gifted students in Math and Science and finished it with honors.

Projects

Master's thesis: Analysis of trends in world cinematography in the early 21st century
2025 [Python, NetworkX]
Analyzed collaboration networks of actors, directors, and production companies across ~1.8M IMDB titles (2001–2024), combining statistical analysis with social network analysis — centrality, community detection, clustering, and assortativity — using Python, NetworkX/igraph, and Gephi. Data collected via the TMDB API and Rotten Tomatoes scraping.

Analysis of scientific production
2024 [Python]
Social network analysis of scientific production in the field of epidemiology and infectious diseases at the Faculty of Medicine, University of Belgrade.

Diabetes detection
2023 [*Python*]
I conducted a diabetes detection project using a database of 100,000 samples, addressing class imbalance with under-sampling and ADASYN techniques. Classification done using KNN, Logistic Regression, and Decision Trees.

Micro-Java Compiler
2023 [*Java*]
Made a simple compiler for Micro-Java. JFlex and Cup tools were used for implementing lexical and syntactic analysis.

Library MEAN web application
2022 [*TS-Angular*]
Web application for library users.
Made with MEAN framework and mongoose database.

Multi-threading kernel
2021-2022 [*C++*]
Thread management subsystem, with time-sharing, context switch, threads, and semaphores. For x86 CPU and RiscV architecture.

Skills

PROGRAMMING: Java, JavaScript, Vue (2/3, Vuex, Pinia, Vuetify), Python, C, C++, React, Angular, VHDL
BACKEND & DATA: REST APIs, JWT authentication, SQL Server, Oracle, PostgreSQL, ArcGIS REST services, ArcPy / Geoprocessing
TOOLS & PRACTICE: Git, Jira, Agile, deployment & production support
LANGUAGE: English, Serbian (native)

Communication

Through non-work experiences, I actively cultivated and enhanced my soft skills, particularly focusing on communication, teamwork, and mentorship. Engaging in various projects allowed me to develop effective collaboration within a team setting